

1. Conditional probability

A survey contains 100 students.

40 students take statistics (S).

24 of those statistics students also take economics (E).

Question: among statistics students, what fraction take economics?

2. Change the reference group

$P(E \mid S) = P(E \text{ and } S) / P(S)$, provided $P(S) > 0$.

$P(E \mid S) = (24/100) / (40/100) = 24/40 = 0.60$.

$P(E \text{ and } S) = 0.24$ is a different quantity.

Conditioning changes the denominator to the specified group.